

Grid Connection Costs as a Barrier to New Generation: Evidence from PJM and Implications for Transmission Policy

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Abstract

Meeting projected growth in electricity demand and climate goals will require building new electricity generators. The grid connection process is seen as a key constraint on this development. We collect new data on grid connection costs for PJM, the largest regional grid operator in the United States. We geographically match these costs to transmission spending to study their determinants. Using regression analysis, we find that these costs, and especially the network upgrade portion, are difficult to predict: generators with similar characteristics can have very different costs. We also find that planned generators with high network upgrade costs are much more likely to be canceled. Finally, prior transmission spending by the grid operator is associated with lower network upgrade costs for connecting generators. These findings emphasize the critical role of transmission capacity in expanding electricity generation capacity.

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1 Introduction

After two decades of less than 1 percent annual growth, US electricity demand is now projected to grow 3 percent per year for 2025–2029 (Wilson, Zimmerman and Gramlich, 2024). This dramatic increase is driven by data centers, manufacturing, buildings, and vehicles. Solar and wind accounted for most new generation capacity in recent years (U.S. Energy Information Administration, 2022, 2023, 2024), raising the possibility that this increased demand can be met without sacrificing climate goals.

Yet, concern is growing that barriers to building new generators will prevent timely investment to meet this demand. Connecting to the electric grid, obtaining the necessary permits, and securing local support are all challenges that can result in project cancellation (Nilson, Hoen and Rand, 2024; Robertson, Rennert and Palmer, 2024). The North American Electric Reliability Corporation concluded in 2024 that more than half of North America was at risk of electricity shortfalls in the next 10 years (North American Electric Reliability Corporation, 2024), and grid operators are asking coal plants to postpone planned retirements (Gheorghiu, 2022; Jao, 2024).

We study how the costs of grid connection impede investment in generation capacity. New utility-scale generators must connect before they begin operation, which requires waiting in a queue for a series of engineering studies to determine the interconnection cost. This cost consists of two components: the point of interconnection (POI) cost, which is the cost of attaching the generator to the grid, and the network upgrade cost, which is the cost of upgrading the network to accommodate its production. In most of the United States, connecting generators are responsible for both types of costs (DeLosa III, Pfeifenberger and Joskow, 2024).

Industry stakeholders have criticized these network upgrade costs and their underlying funding model. They argue that the system places a disproportionate burden on new generation (Caspary et al., 2021; Armstrong et al., 2024) and that inadequate levels of centrally planned investment in transmission infrastructure result in overly high network upgrade costs. In PJM, transmission investment, which is paid for by electricity consumers rather than by connecting generators, has been roughly stable over the last decade even as requests for grid connection have grown.

This debate has become more pressing in recent years due to rising interconnection costs and their correlation with project cancellation: Gorman et al. (2024) compiled the first systematic data on these costs and found that network costs have risen dramatically over the last decade, an increase driven by costs for projects that withdraw from the queue. This finding highlights the need for a deeper understanding of these costs and their determinants.

We address this gap using new data on grid connection costs and transmission investment in PJM, the largest US regional transmission grid operator by customers served. PJM is particularly interesting because rising demand relative to new generation coming online contributed to record-high capacity market prices in 2025, prompting nine governors to suggest that their states were losing confidence in the grid operator (Penn, 2025). These PJM data include cost estimates from every engineering study a project receives. We can therefore analyze the relationship between costs and project cancellation at each stage in the grid connection process and describe how estimated costs for the same generator evolve across engineering studies.

This paper provides three key findings that have implications for transmission policy. First, we show that network upgrade costs are unpredictable. Generators with similar sizes, fuel types, and locations can face very different network upgrade costs. The estimated cost for the same generator can also change significantly while it waits in the queue, a pattern that holds for both renewables and other generators.

Second, we find that high network upgrade costs lead to project cancellation. Generators with network upgrade costs in the second study are 59 percent more likely to withdraw from the queue than those without. This cancellation effect is larger for renewables: a nonzero second-study network upgrade cost raises the probability of withdrawal by about 0.30 for wind and solar projects, compared with 0.18 for other generator types. Among generators with nonzero costs, doubling the cost increases the withdrawal probability by 8 percent.

Third, we find that network upgrade costs are lower after transmission investment by the grid operator. New generators in areas with high recent transmission spending are 50–68 percent less likely to face high network upgrade costs, consistent with connecting generators benefiting from infrastructure paid for by electricity consumers.

Taken together, these findings suggest that the current generator-pays model for network upgrades, combined with a lack of proactive transmission investment, impedes the expansion of generation capacity. Section 7 compares this model to alternative cost-allocation regimes used elsewhere that shift costs away from generators, thus lowering the cost barrier we document but weakening generators’ incentives to site in low-cost areas of the network.

This paper contributes to the literature on renewable energy deployment. Recent work has documented various barriers to investment in renewable energy, including permitting challenges (Nilson, Hoen and Rand, 2024; Liu, 2025) and local opposition (Stokes et al., 2023). For grid connection, Johnston, Liu and Yang (2026) study how queue design affects delays and completion, while Gorman et al. (2024) document rising interconnection costs nationwide. We extend this literature by analyzing how a project’s network upgrade costs vary across multiple engineering studies and cause withdrawal from the queue.

We also contribute to the literature on transmission investment and its economic value.¹ Recent evidence shows that large transmission expansions spur renewable energy entry (Gonzales, Ito and Reguant, 2023; Doshi, 2024). We extend this evidence by documenting that even moderate levels of transmission investment are associated with lower network upgrade costs for connecting generators—costs that we find cause project cancellation.

2 Setting and Data

2.1 The PJM Interconnection Process

PJM operates the transmission system serving about 65 million people across 13 states and the District of Columbia (PJM, 2021). A proposed generator must proceed through PJM’s interconnection queue, which determines whether the project can reliably connect to the grid and identifies the transmission upgrades required to accommodate it. As a project moves through the queue, its developer receives increasingly precise estimates of these upgrades and their costs.

2.1.1 Engineering Studies

For most projects, PJM requires three engineering studies: feasibility, system impact, and facilities, which we refer to as the first, second, and third studies.

The three studies serve distinct engineering purposes and progressively refine the information available to developers. The first study provides a preliminary assessment of technical feasibility and an initial estimate of the interconnection costs. The second study evaluates the project’s effects on system reliability and stability, identifies the specific network upgrades for which the project is responsible, and produces the project’s first estimate of its allocated network upgrade cost. The third study develops detailed engineering designs and the most refined cost estimate, which forms the basis for the project’s interconnection agreement. Because each study reflects the state of the grid and queue at the time it is conducted, a project’s estimated costs can change substantially across studies.

Not all projects require three studies. If PJM determines that a project needs no network upgrades and affects no nearby projects, it may issue a single combined feasibility-and-impact study, which we classify as the first study, in place of the first two.² This expedited path is

¹See, e.g., Wolak (2015); Davis and Hausman (2016); Ryan (2021); Fell, Kaffine and Novan (2021); LaRiviere and Lyu (2022); Yang (2022); Hausman (forthcoming), and Ham, Kay and Hausman (2025).

²Less commonly, a project may receive a combined impact-and-facility study or a combined feasibility-impact-and-facility study, which we classify as the third study.

used primarily for projects below 20 MW. Eighteen percent of projects in our sample receive some form of combined study rather than three separate studies.

2.1.2 Interconnection Costs and Allocation

A generator’s interconnection costs have two main components. Point-of-interconnection (POI) costs cover the equipment needed to physically connect the project to the grid; connecting generators bear these costs in essentially every system. Network upgrade costs cover broader reinforcements to the transmission system required to accommodate the project’s output. For these, PJM follows a generator-pays cost-causation framework: the project that triggers the need for an upgrade is assigned responsibility for its cost. As we discuss in Section 7, this allocation of network upgrade costs is a key institutional feature that varies across systems.

PJM reports different cost measures across studies. The first study provides a POI cost estimate and a combined network upgrade cost, which we refer to as the “combined cost.” This is the total estimated cost of all network upgrades that a project may be required to help fund, before its specific share has been determined. The second and third studies provide estimates of the POI costs, the combined cost, and the network upgrade cost, which is the cost of the upgrades that PJM has determined the project is responsible for funding. The combined cost, which is typically much larger, serves as an early, approximate signal of the network upgrade cost.

When multiple projects jointly trigger a network upgrade, PJM forms a cost-share group rather than assigning the full cost to a single project. Each project in the group is charged in proportion to its contribution to the overload that the upgrade addresses. Because cost shares depend on which projects are in the group, a project’s network upgrade cost can change when other members exit the queue. Appendix A.1 works through an example of how these costs are reported and shared.

2.1.3 Interaction with Large Load Interconnection

Load interconnection interacts with generator interconnection through PJM’s load forecast, which helps determine the network conditions modeled in generator interconnection studies. How load growth affects a given generator’s network upgrade costs depends on where the growth occurs. Load growth near the generator’s point of interconnection can absorb its output locally and reduce the need for upgrades. By contrast, growth in distant locations can create congestion on the transmission paths the generator would use, increasing the need for upgrades. Because each engineering study reflects the load forecast at the time

it is conducted, forecast revisions can alter a generator’s estimated network upgrade costs; Section 4 examines this channel empirically. PJM’s 2022 load forecast, for example, projected approximately 5,700 MW of data center load growth in the Dominion zone by 2037; the 2025 forecast projected more than 20,000 MW over the same horizon (PJM Interconnection, 2022, 2025).

Much of PJM’s projected load growth comes from large loads, which are facilities with substantial electricity demand. These loads have historically connected to the grid through their local distribution utility rather than through a formal interconnection queue. The process for connecting large loads is under reconsideration: in 2026, FERC directed PJM and the other regional transmission organizations to justify or revise how large loads interconnect and pay for network upgrades (Federal Energy Regulatory Commission, 2026a).

2.2 New Data on Grid Connection Costs

We leverage new interconnection cost data for PJM. Our analysis uses hand-collected interconnection cost data from engineering studies for all projects that entered the PJM interconnection queue from 2008 to 2020. Many of these projects are still in the queue, and our data on their studies go through 2024. Hand-collection was necessary because interconnection cost data are not collected by the US Energy Information Administration or made easily accessible by regional transmission grid operators (Gorman et al., 2024).

We collect the two main components of interconnection costs, POI costs and network upgrade costs, separately. We also collect the combined cost, the total cost of the network upgrades to which a project may need to contribute. Table 1 reports summary statistics for these data. Most requests for interconnection have POI costs; those that do not are typically capacity increases at existing plants. Network upgrade costs are rarer; for example, 27 percent of second-study observations have nonzero network costs. For more detail on data collection for these variables, see Appendix A.1.

Table 1 also summarizes two indicators related to a project’s network upgrade costs. The *network allocation* indicator equals one if the study reports an explicit network upgrade cost for the project, even if the reported cost is zero. The *cost-share group* indicator equals one if the study lists the other generators with which the project shares an upgrade cost. Network allocations appear for 59 percent of second-study observations and 63 percent of third-study observations, while cost-share groups are identified for 17 percent and 26 percent, respectively.

The bottom rows of Table 1 describe project characteristics. About a fifth of requests are capacity increases at existing plants and around 70 percent are wind or solar projects.

Average project size rises across the three studies (98, 106, and 157 MW). This pattern reflects changing sample composition rather than growth in individual projects: among projects observed in all three studies, average size is nearly constant. One reason the composition shifts is the expedited path described above, where smaller projects are less likely to require all three studies.

These project-level data cover a period of rapid growth in the PJM interconnection queue. Appendix Figure B1 shows that entries to the queue were roughly five times as high in 2020 and 2021 as in 2011 and 2012. The increase in requested capacity was also large: from roughly 20 GW per year to more than 60. This increase corresponded with a shift in the composition of requests away from natural gas and toward solar, wind, and batteries. Grid operators across the United States and Europe saw similar increases in requests and shifts toward renewable generators (Gorman et al., 2024; European Commission, 2025).

These data are complementary to the nationwide data on US grid connection costs from Gorman et al. (2024). Our data from one grid operator include the cost from every engineering study, while the data in Gorman et al. (2024) come from many grid operators and include the cost from the terminal engineering study. This terminal study may be the first, second, or third study depending on when the project left the queue. Observing data from many regions allows Gorman et al. (2024) to compare costs in regions with different resource mixes and policies whereas our data allow for a careful quantification of the relationship between costs and withdrawal decisions at each stage in the process. Comparing generators at the same stage is important because they select out of the queue based on their estimated interconnection costs. In Johnston, Liu and Yang (2026), we also use these data to study the PJM queue, but that paper focuses on queue congestion and reforms to queue design.

Table 1: Summary Statistics for PJM Interconnection Queue Data

	First Study		Second Study		Third Study	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Network upgrade cost (\$/kW)	-	-	77.43	243.66	13.96	68.92
... Fraction nonzero	-	-	0.27	0.45	0.18	0.38
Combined cost (\$/kW)	931.68	3,073.11	707.15	1,815.28	61.90	134.38
... Fraction nonzero	0.41	0.49	0.21	0.41	0.04	0.20
POI cost (\$/kW)	104.88	221.92	82.82	167.07	102.37	168.94
... Fraction nonzero	0.63	0.48	0.71	0.45	0.88	0.33
Cost-share group	-	-	0.17	0.38	0.26	0.44
Network allocation	-	-	0.59	0.49	0.63	0.48
Size (MW)	97.75	196.45	105.65	190.71	157.49	257.86
Capacity increase at existing plant	0.21	0.41	0.23	0.42	0.15	0.36
High voltage (POI voltage > 115kV)	0.45	0.50	0.51	0.50	0.57	0.49
Wind/solar	0.68	0.47	0.70	0.46	0.73	0.44
N	4,133		2,472		823	

The sample is all generators that entered the queue from 2008 to 2020. We collect data from the final version of each study. Costs are in 2023 dollars per kW. Fraction nonzero is the share of observations with cost estimates of nonzero. Cost-share group is a binary indicator equal to 1 if the study lists other generators with which the project shares network upgrade costs, and 0 otherwise. Network allocation is a binary indicator equal to 1 if the study reports an explicit network upgrade cost for the project, and 0 otherwise. Size (MW) refers to the maximum of the generator's requested energy or requested capacity in MW. Capacity increase at existing plant is a binary indicator for a request to increase maximum generating capacity at an existing power plant; the size for these requests is the incremental capacity, not the total plant capacity. High Voltage is a binary indicator for a point of interconnection with voltage greater than 115 kV. Wind/Solar is a binary indicator for a wind or solar project.

3 Network Upgrade Costs Are Difficult to Predict

We first examine how network upgrade costs change over time and vary with project characteristics. We also compare them to POI costs. All costs are reported in 2023 dollars.

Figure 1 shows that network upgrade cost estimates increase over time for the second but not the third study. In the second study, later cohorts of entrants have both a higher probability of network upgrade costs (Panel A) and higher such costs (Panel C). This increase

is present for both wind and solar (blue) and other fuel types (red). We do not see a similar increase in second-study POI costs (Appendix Figure B2). In the third study, the estimated network upgrade costs are low (D), and, if anything, the share of projects with network upgrade costs is decreasing over time (B). While there are differences across fuel types in the probability of having a nonzero cost, average third-study costs across all projects are low for both wind and solar (\$13/kW) and other generators (\$11/kW).

We next map second-study network upgrade costs and find that geographically adjacent projects can have very different network upgrade costs. Figure 2 maps these costs separately for generators that entered the queue in 2011–2018 (Panel A) and 2019–2020 (Panel B). Costs tend to be higher along the coasts, but most of the variation is local rather than regional. This dispersion is not an artifact of project composition: it persists when we map wind and solar separately from other types of generators (Appendix Figure B3) and when we partial out size, fuel, their interactions, transmission zone, voltage, whether the project is a capacity increase at an existing plant, and queue year (Appendix Figure B4). POI costs vary within narrow geographies as well (Appendix Figure B5).

We also use regression analysis to examine which project characteristics are most predictive of second-study network upgrade costs, and find that observable characteristics explain little of the variation. Appendix Table B1 shows that smaller projects are less likely to have network upgrade costs but, if they do, have higher costs per kW. We do not find that network upgrade costs are higher for renewable generators than for other fuel types, at least on a cost per kW basis. Across all specifications, observable characteristics explain no more than 40 percent of the variation in network upgrade costs. Observable characteristics explain more of the variation in POI costs but still less than 50 percent. We also do not find that network upgrade costs are higher when the transmission owner (TO), which conducts the engineering studies in conjunction with PJM, owns incumbent generators whose revenues may be reduced by the entry of new generators (Appendix A.2 and Appendix Figure B6).

4 Network Costs Can Change Across Studies

We next describe how network upgrade costs for the same generator evolve across studies. There are two explanations for the above finding that average network upgrade costs are much higher for the second study than they are for the third study. One is that these costs fall systematically across studies. Another is that generators with high costs in the second study exit before they reach the third. The first explanation implies that network upgrade costs for the same generator should, on average, fall across studies, but the second does not.

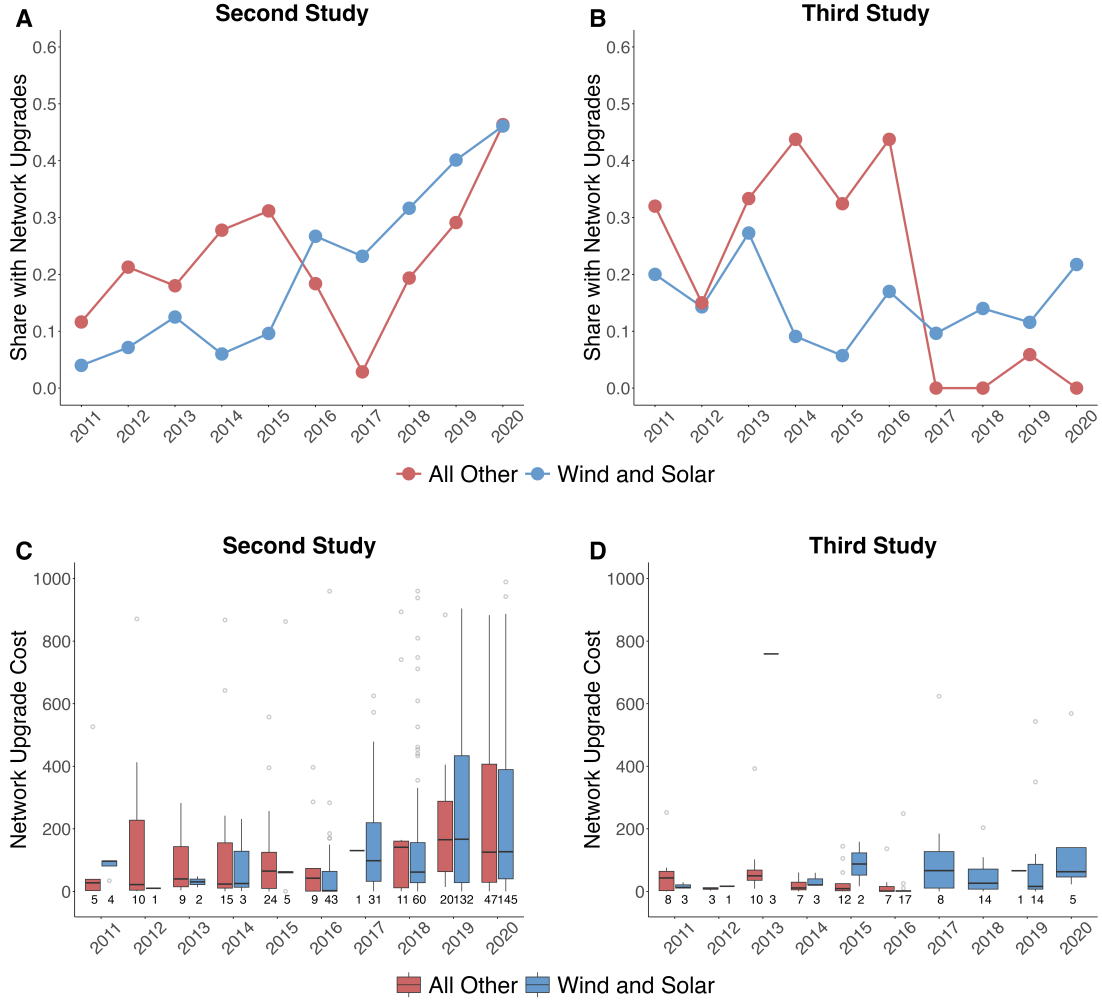


Figure 1: **Network Upgrade Costs by Year of Queue Entry and Fuel** (A) Share of projects with nonzero network upgrade costs in the second study. (B) Share of projects with nonzero network upgrade costs in the third study. (C) Box plot of second-study network upgrade costs in \$/kW for projects with nonzero costs. (D) Box plot of third-study network upgrade costs in \$/kW for projects with nonzero costs. Numbers below bars in C and D denote the number of projects with nonzero costs.

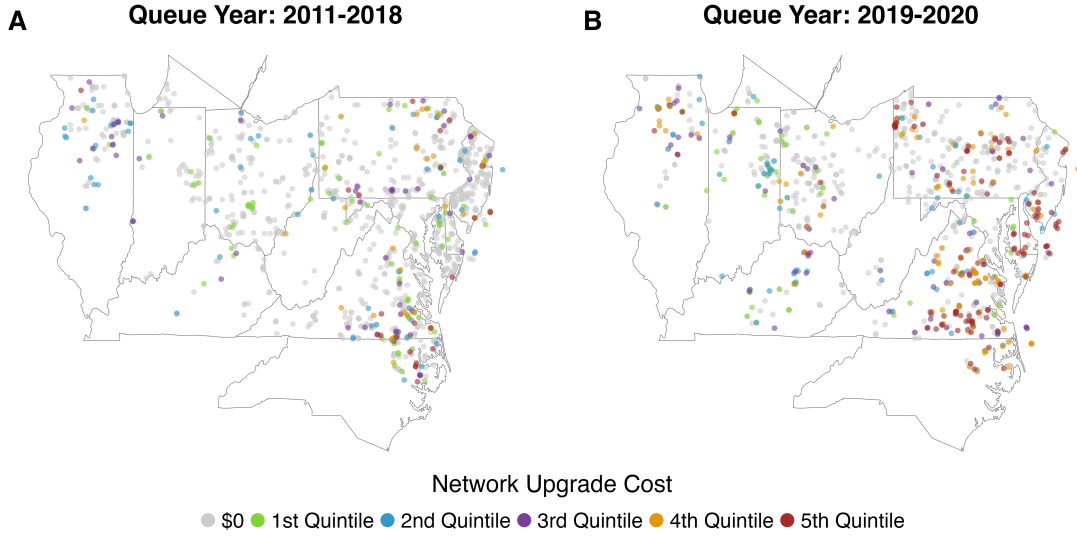


Figure 2: **Network Upgrade Costs by Year of Queue Entry and Location** Estimated network upgrade costs from the second study for the 1,144 projects that entered the queue in 2011–2018 (**A**) and the 922 projects that entered the queue in 2019–2020 (**B**). Network upgrade costs are grouped into five bins defined by the 20th, 40th, 60th, 80th, and 100th percentiles, in addition to a zero-cost group. The corresponding cutoffs are 14, 68, 170, and 512 \$/kW.

We find that a generator’s estimated network upgrade cost is equally likely to increase or decrease across studies, and large changes can occur. Most generators (74 percent of wind and solar, 62 percent of other) that make it to the third study had estimates of zero for both studies. For generators with at least one nonzero network upgrade cost, the median change from the second to the third study is $-\$0.03/\text{kW}$, the mean is $-\$28/\text{kW}$, and the standard deviation is $\$191/\text{kW}$. This pattern suggests that the difference in average cost for the two studies is due to generators with high costs in the second study withdrawing before the third study. Figure 3, panels A and C show that this pattern holds for both renewables and other fuel types. Appendix Figure B8 shows that POI costs often change substantially as well.

We next investigate why network costs for the same generator change across studies. A leading explanation is that network conditions change between studies. Projects that share network upgrade costs may be especially prone to this type of cost change, and costs can either increase or decrease. If a generator in a cost-share group withdraws and the upgrade is still needed, the departing generator’s share is redistributed to the remaining members, raising their costs. If instead the withdrawal makes the upgrade unnecessary, the upgrade is dropped and costs fall for the remaining members. Changing demand can also alter network conditions. Because each study models the network under the load forecast at the time, forecast revisions between studies can change the required upgrades.

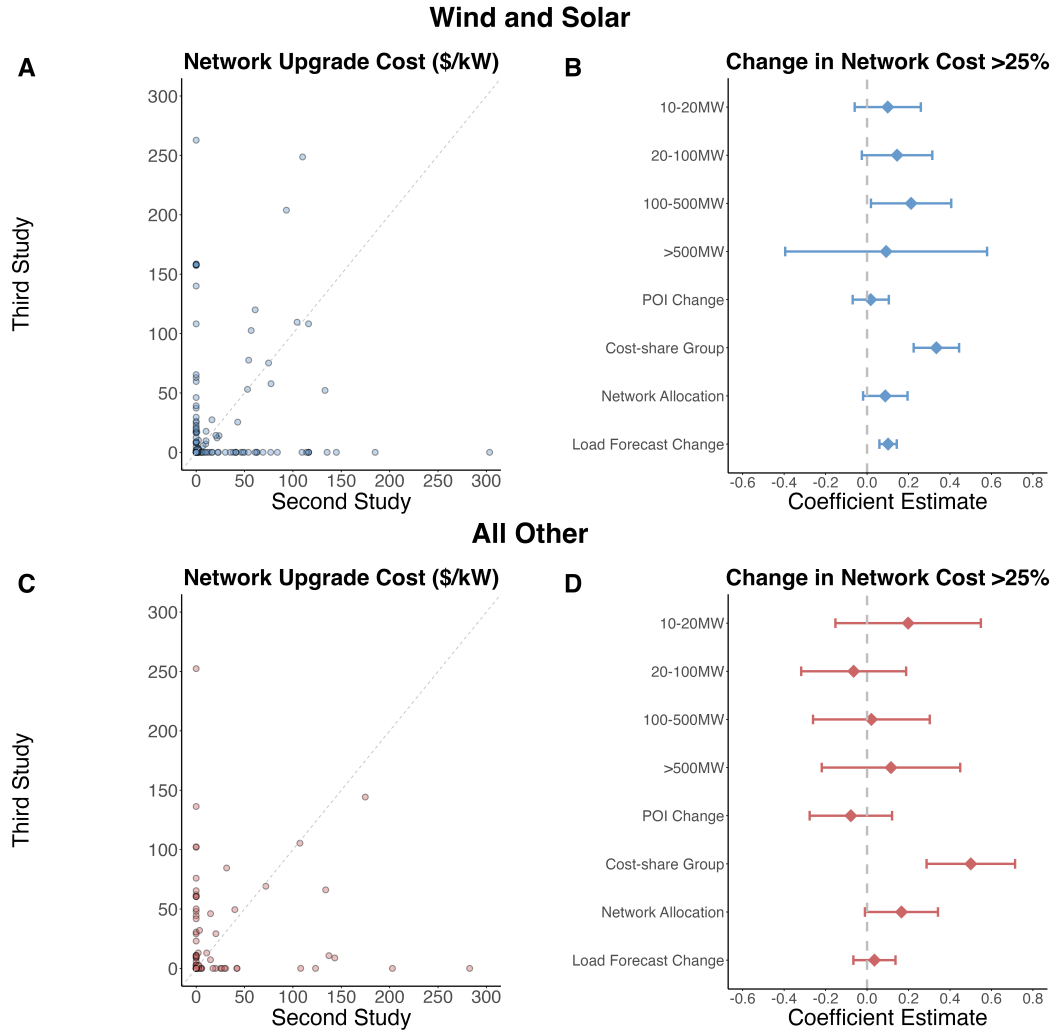


Figure 3: Comparison of Second- and Third-Study Network Upgrade Costs for the Same Project and Predictors of Large Changes, by Fuel Type. Panels (A) and (C) plot third-study versus second-study network upgrade costs for projects observed in both studies, separately for wind and solar (Panel A: 573 projects, of which 423 (74 percent) have zero network upgrade costs in both studies; 19 projects with at least one cost greater than \$300/kW are excluded from the plot) and all other fuel types (Panel C: 199 projects, of which 123 (62 percent) have zero network upgrade costs in both studies; 3 projects with at least one cost greater than \$300/kW are excluded from the plot). Panels (B) and (D) report estimated coefficients from a linear probability model predicting the probability of a network upgrade cost change greater than 25 percent between the second and third study, estimated separately for wind and solar (Panel B, sample mean = 0.21) and all other fuel types (Panel D, sample mean = 0.36). Size coefficients are relative to an omitted bin for 0–10 MW. POI Change indicates the POI changed between the second and third studies. Cost-share group is an indicator for a second study that lists other generators that a project shares network upgrade costs with. Network allocation is an indicator for an explicit network upgrade cost in the second study; this cost may be \$0. Load Forecast Change is the absolute change, measured in GW, in the five-year-ahead PJM peak-load forecast for the project’s transmission owner between the second and third studies. The model controls for fuel type, whether the project is a capacity increase at an existing plant, year of entry into the queue, state, transmission owner, and voltage.

We find that the biggest predictor of a large change is if the generator shares costs. Figure 3 panel B and panel D show select coefficients from regressing an indicator for a large change in network upgrade costs (>25 percent) between the second and third study on project characteristics, separately by fuel type. Among projects that reach the third study, 27 percent of renewable projects and 24 percent of other projects have a second study that identifies their cost-share group. For renewable projects, cost sharing is associated with a 33 percentage point increase in the probability of a large cost change, a 157 percent increase relative to the mean of the dependent variable (0.21). For other projects, the corresponding increase is 50 percentage points, a 138 percent increase relative to the mean (0.36).³ We also find that changes in load forecasts are predictive of large changes for renewables: a one-standard-deviation increase raises the probability by 9.6 percentage points, or about 45 percent of the sample mean. These variables are less predictive of changes in POI costs (Appendix Figure B8, panels B and D).

5 High Costs Result in Project Cancellation

We next use regression analysis to quantify how interconnection costs affect the probability that projects are canceled. We conduct the analysis separately for the first, second, and third studies. The dependent variable is an indicator for withdrawing from the queue after receiving a study, and the two regressors of interest are the two components of interconnection costs. For the first-study regressions, we use the combined cost as our measure of the network upgrade cost. Among projects that reach the second study, the correlation between the first-study combined cost and second-study network upgrade cost is 0.33.

Figure 4 plots the estimated coefficients, with each panel reporting estimates for a different sample. Panels A, C, and E show the extensive margin: the effects on withdrawal of having nonzero interconnection costs for all projects that receive the first, second, and third studies. Panels B, D, and F show the intensive margin: the effects of log costs on withdrawal among projects with nonzero estimates for both types of costs, estimated separately for each of the three studies. We report estimates from four models that vary from no controls (green triangles) to rich controls for project characteristics (orange circles). Re-estimating the withdrawal specifications on the wind and solar subsample yields the same pattern (Appendix Figure B9).

Projects with high interconnection costs are more likely to be canceled. In the first study, we do not see that projects with nonzero POI or combined costs are more likely to withdraw

³Appendix Figure B7 reports results for analogous models of large cost increases and decreases. Cost-sharing is more predictive for large decreases while a network allocation is more predictive for large increases.

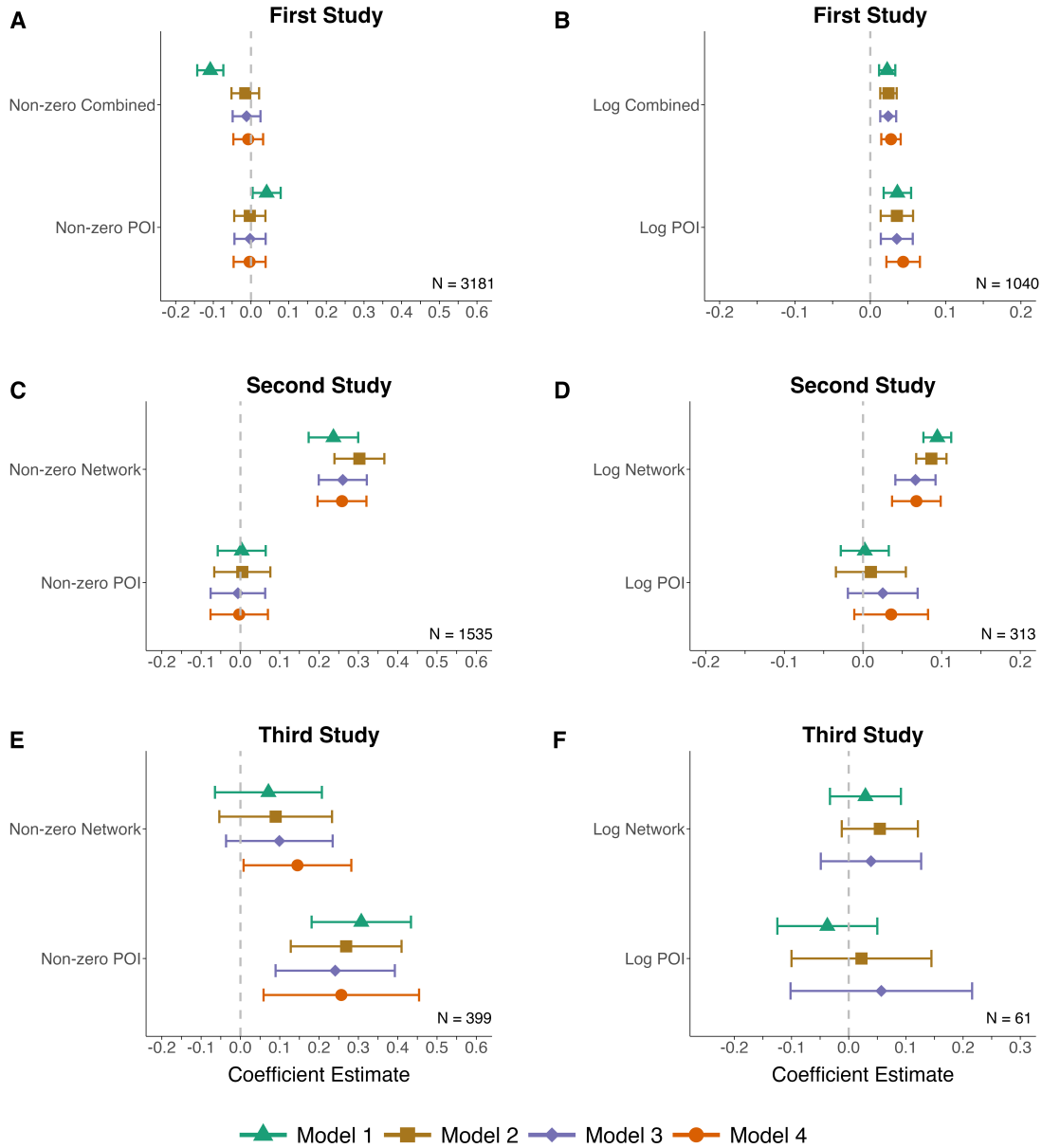


Figure 4: **Effect of Interconnection Costs on Probability of Withdrawal.** Coefficient point estimates (triangles, squares, diamonds, and circles) and 95 percent confidence intervals from a linear probability model. Projects that entered the queue 2011–2020, active projects excluded. Dependent variable is an indicator for withdrawing from the queue before the next study (first and second study) or beginning operation (third study). The means of the dependent variables are 0.285 (A), 0.238 (B), 0.437 (C), 0.607 (D), 0.479 (E), and 0.557 (F). Model 1 (green triangles) has no controls, Model 2 (brown squares) controls for project size and fuel type, Model 3 (purple diamonds) also controls for year of entry into the queue, and Model 4 (orange circles) also controls for state, transmission owner, and voltage. Panels A, C, and E plot coefficients on indicators for a nonzero combined or network upgrade cost (the extensive margin), estimated on all projects that receive the first, second, and third study, respectively. Panels B, D, and F plot coefficients on the log of the cost (the intensive margin) among projects with positive costs in that study. The samples are accordingly: all projects with a first study (A), projects with positive first-study combined and POI costs (B), all projects with a second study (C), projects with positive second-study network and POI costs (D), all projects with a third study (E), and projects with positive third-study network and POI costs (F).

from the queue than those with costs of zero (Panel A). But among projects with nonzero costs, those with higher POI costs are more likely to withdraw: the estimated coefficient from Model 4 implies that a doubling of POI costs increases the probability of withdrawal by 0.03, or 13 percent at the mean withdrawal rate for this sample (Panel B). Projects with a higher combined cost, an indication of future network upgrade costs, are also more likely to withdraw.

We estimate large effects of second-study network upgrade costs on withdrawal (Panels C and D). The estimates from Model 4 imply that, compared to projects without such costs, those responsible for network upgrades are 26 percentage points, or 59 percent, more likely to withdraw before the third study. For the sample with network upgrade costs, doubling the cost increases the probability of withdrawal by 0.05, or 8 percent. We do not find that projects with higher second-study POI costs are more likely to withdraw.

This pattern is consistent with Seel et al. (2023), who find that withdrawn projects in PJM have systematically higher interconnection costs than those that reach operation and that this difference is driven by network upgrade costs. Their evidence comes from unconditional cost averages by request status, whereas our estimates compare projects at the same study stage and hold fixed characteristics such as size, fuel type, and location. Johnston, Liu and Yang (2026) also find that high interconnection costs increase the probability of withdrawal, but do not separately estimate the effects of network upgrade costs and POI costs.

The extensive-margin withdrawal response to second-study network upgrade costs is larger for renewable projects; we find no corresponding difference on the intensive margin. In a pooled specification that interacts fuel type with the nonzero-cost indicator, incurring a nonzero second-study network upgrade cost increases the probability of withdrawal by 0.30 for wind and solar projects, compared with 0.18 for other generators. Both estimates are statistically significant, as is the difference between them ($p = 0.04$). By contrast, we find no fuel-type difference among projects that incur a positive cost: the estimated coefficient on log second-study network upgrade cost is 0.07 for wind and solar projects and 0.08 for other generators ($p = 0.86$ for the difference).

Taken together, the pattern of estimates across studies is consistent with the interpretation that interconnection cost estimates contain new information that affects cancellation. A developer likely knows whether a project will have POI costs but learns more about how large they will be in the first study, which may explain the intensive-margin effects we find in Panel B. Developers must wait until the second study to learn whether a project is likely to have network upgrade costs and how large they might be. At that point, we find large impacts of these costs on withdrawal from the queue. These effects are consistent with network upgrade costs being difficult for developers to predict.

These estimates likely understate the causal impact of interconnection costs on project cancellation. A developer would plan a project that it expects to have high costs only if the project were otherwise especially good. This behavior creates a positive correlation between unobserved project quality and interconnection costs, biasing the coefficients on these costs in the withdrawal regression toward zero. More generally, these estimates could reflect the effects of other variables that are correlated with interconnection costs, but, reassuringly, we find similar estimates as we add more control variables to the model.

The magnitude of these costs helps explain why they are so strongly associated with withdrawal. Recent realized installed costs for utility-scale solar and wind (land-based), the technologies that dominate the queue, are roughly \$1,370 to \$1,500 per kW (Wiser et al., 2023; Seel et al., 2024). Relative to this benchmark, second-study total interconnection costs average about \$160 per kW, or 11 percent of the cost of building the project, of which network upgrade costs account for about 6 percentage points. These averages mask wide variation across projects. Among the 27 percent of projects that trigger network upgrades, these costs average about \$296 per kW, around 20 percent of construction cost, and reach about \$870 per kW at the 90th percentile. For these projects, grid connection costs raise up-front capital costs by more than half.

6 Costs Are Lower After Transmission Investment

Despite much discussion about the benefits of interregional transmission capacity (Millstein et al., 2024; Hausman, forthcoming), most US spending on transmission is intraregional (Pfeifenberger et al., 2021). Grid operators plan transmission investments through a regional transmission planning process. We next summarize data on this planned transmission investment for PJM and look at the correlation between these investments and network upgrade costs.

Through its annual Regional Transmission Expansion Plan (RTEP) process, PJM approves \$1–6 billion per year in investments. Although demand for grid connection in PJM has grown substantially over the last 10 years, transmission investment has been roughly stable (Appendix Figure B10). One reason may be that the planning process has historically not considered generators waiting in the queue until they reach the third study (Lieberman, 2021).

This RTEP spending is divided about equally between baseline and supplemental projects. Baseline projects are necessary to maintain “system reliability, operational performance, or economic criteria”; supplemental projects are not (Tesfa, 2023). The latter, also known as “local projects,” primarily address aging infrastructure and customer service needs, such as

connecting new loads (Dominion Energy, 2025). They are planned by the local transmission owner, receive less oversight than baseline projects, and may be less cost-effective (Macey and Mays, 2024).⁴ Unlike the network upgrades made by connecting generators, both baseline and supplemental projects are paid for by load (electricity consumers) via transmission rates. Appendix Table B2 shows that baseline and supplemental projects are similar in terms of cost, number of substations the work is performed on, and time to operation. The average voltage for baseline projects is about 30 percent higher than that of supplemental projects.

We next study the relationship between this spending and network upgrade cost estimates for connecting generators. We regress an indicator for whether a project faces a high network upgrade cost ($> \$100/\text{kW}$) in the second study on the total cost of nearby RTEP transmission projects that have entered operation. Specifically, we include RTEP projects that went into service between three years before and one year after the issue date of the second study. Because the transmission grid is a complex network, we present results for four definitions of nearby spending, acknowledging that all are imperfect. The first three measure aggregate transmission spending at substations within 10km, 20km, and 50km of the POI of the connecting generator. The fourth is the aggregate transmission spending by regions that we construct based on local wholesale electricity prices. Specifically, we use hierarchical clustering to group substations into 50 regions based on location and similarity in locational marginal prices. For comparability across models, we bin spending by the percentiles of the distribution for that definition, and coefficients are interpreted as effects relative to the omitted group, which is spending levels below the median for that definition. For more detail, see Appendix A.3.

We find that generators in areas with high levels of recent transmission investment are less likely to have high network upgrade costs. Figure 5, Panel A shows a generally negative relationship between total transmission spending and the probability of a high network upgrade cost. This negative effect is statistically significant in the top two spending bins for spending within 50km, and for the 75th to 90th percentile bin for spending in the same region. For these two highest levels of geographic aggregation, our estimates imply that, compared to generators in areas with below-median investment, those in areas above the 75th percentile are 8–11 percentage points less likely to have high network upgrade costs. These effects represent a 50–68 percent reduction relative to the mean. Estimating these

⁴Macey and Mays (2024) note that, in the last decade, regional transmission investment has shifted away from baseline projects and toward local projects. They argue the result is a “piecemeal process in which lines are built in response to one-off needs and in which incumbents steer investment towards projects that protect their financial interests but do not provide the most cost-effective approach to meeting the country’s transmission needs.”

models separately by fuel type yields a similar negative relationship for both renewables and other generators.

Panel B further disaggregates transmission spending by category and shows that these results are not driven by baseline investment. This unexpected finding may be because the baseline and supplemental projects in PJM are similar in size. The vast majority of PJM’s baseline projects are also for reliability rather than congestion relief and thus may be less likely to increase network capacity. Only 1.3 percent of baseline projects address congestion relief, compared to 95 percent that are driven by reliability needs.

We caution that our transmission spending variables are imperfect measures of the investment relevant to each generator. Measurement error from temporal and spatial mismatch would bias our estimates, likely toward zero. Transmission investments take years to plan and build, and their benefits may not be realized in our observation window. The transmission investment that affects a generator’s costs also need not be located near it: because power flows across the entire network, investment at distant substations can relieve the constraints that would otherwise require upgrades at a generator’s expense. Our measures based on spending within 10, 20, and 50 km of the POI may miss this investment. This limitation motivates the region-based measure, which groups substations by both wholesale price similarity and geographic proximity, although it, too, is an imperfect proxy.

Entry induced by transmission investment may also bias our estimates toward zero, a concern our control for prior entry only partly mitigates. Developers may disproportionately enter locations with new transmission, competing away its beneficial effects (see Gonzales, Ito and Reguant (2023) and Doshi (2024) for evidence that large-scale transmission expansions spur entry). All models include the log of the MW entering the queue in the corresponding geographical area in the prior three years, and we find a statistically significant positive effect of entry on network upgrade costs.

7 PJM in Context

Although our focus is on PJM, these results likely generalize to most of the United States. Other research has found that rising network upgrade cost estimates are ubiquitous (Gorman et al., 2024), and other grid operators also require multiple engineering studies and share upgrade costs across generators in the queue. The transmission planning process in PJM is similar to that of other grid operators, though PJM plans fewer of the economic and public policy projects that may be especially helpful for integrating new generators (U.S. Department of Energy, 2023).

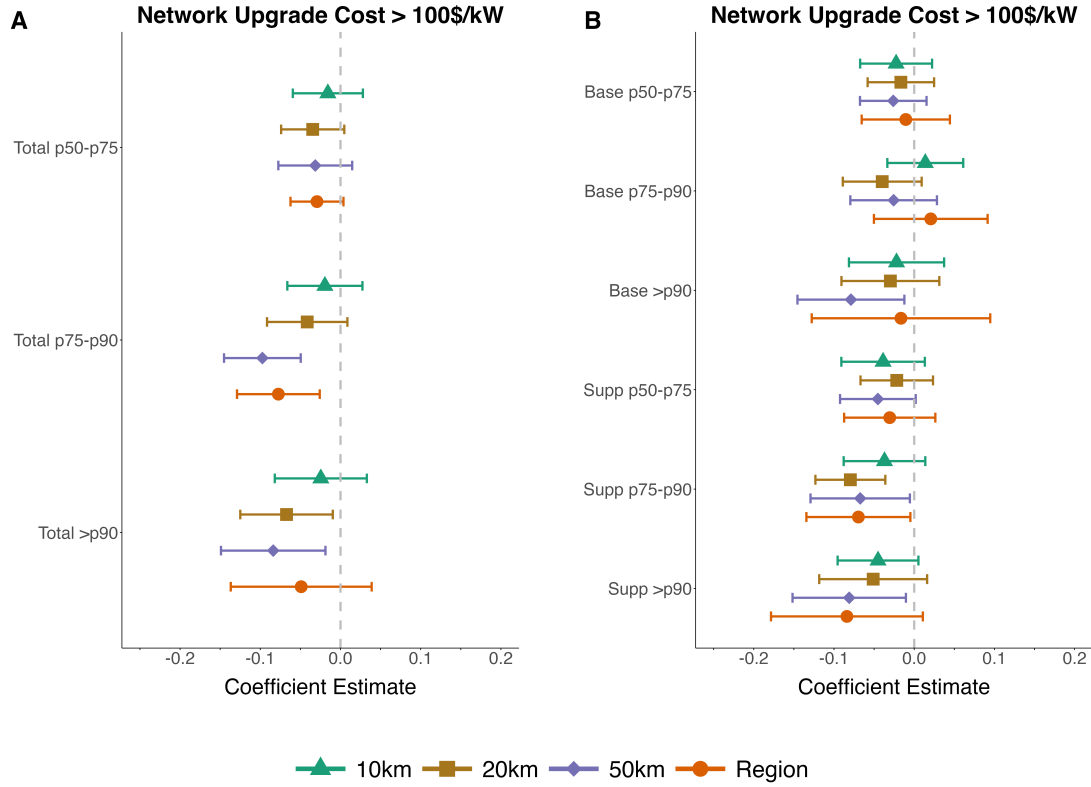


Figure 5: **Relationship Between PJM Transmission Spending and Network Upgrade Costs.** Coefficient point estimates (triangles, squares, diamonds, and circles) and 95 percent confidence intervals from linear probability models. Sample is projects that entered the queue in 2011–2020. Dependent variable is an indicator for a second-study network upgrade cost greater than \$100/kW (mean = 0.15). Independent variables are binned indicators for recent Regional Transmission Expansion Plan transmission spending. Each model corresponds to a different definition of nearby spending: 10km (green triangles), 20km (brown squares), 50km (purple diamonds), or regions constructed based on similarity in locational marginal prices (red circles). Spending is grouped into percentile bins: 50–75th, 75–90th, and 90–100th; the omitted category is spending below the median. Panel A reports results from regressions with three bins based on overall spending. Panel B reports results for regressions with six bins, three for baseline spending and three for supplemental spending. All specifications control for project size, fuel type, whether the project is a capacity increase at an existing plant, load forecast, transmission owner, queue year, and state. They also control for the log of the MW entering the queue in the past three years in the geographic area matching the area used for the transmission spending variables. Standard errors clustered by substation for the 10km, 20km, and 50km models and by region for the Region model.

Some U.S. grid operators differ from our empirical context in how they handle network upgrade costs. ERCOT does not require connecting generators to pay for them: new generators interconnect on the existing system, and ERCOT curtails generation when the transmission system cannot accommodate it. CAISO requires generators to pay upfront based on their contribution to required upgrades, similar to PJM, but it later reimburses generators for some of these costs, with the funds for repayment drawn from charges on load.⁵ ISO-NE follows a similar approach: generators pay upfront for required network upgrades but are reimbursed via credits against their transmission bills within five years of commercial operation, with the funds for these credits drawn from transmission charges paid by load (ISO New England Inc., 2025).

Outside the United States, a common model is to socialize the cost of network upgrades across all users, with generators paying only direct connection costs. Upgrade costs are typically recovered via the transmission rates paid by electricity consumers. This approach is used in most of Europe and in Australia (Agency for the Cooperation of Energy Regulators, 2025; Australian Energy Market Commission, 2017).

The choice between these cost-allocation models involves a tradeoff. In generator-pays models, generators internalize the transmission impact of their siting decisions, but the resulting network upgrade costs are high and uncertain—a barrier to new generation, as our findings show. Socialized-cost models remove this barrier, but weaken locational signals: generators do not internalize the transmission impact of their siting decisions, concentrating projects in resource-rich locations and inflating the total transmission needed to deliver their output.⁶ The Canadian province of Alberta, for example, is now shifting network upgrade costs onto connecting generators after concluding that weak incentives to site in low-cost areas had inflated transmission costs (Government of Alberta, 2024; Alberta Ministry of Affordability and Utilities, 2023).

In theory, the two models err in opposite directions. Generator-pays models can lead to underinvestment: a connecting generator pays for upgrades whose benefits accrue partly to others, so too few upgrades are built. Socialized-cost models can lead to overinvestment, and in the wrong places: generators’ siting choices drive which upgrades are needed, but

⁵CAISO divides network upgrades into reliability network upgrades, which address reliability violations, and delivery network upgrades, which enable output to count toward resource adequacy. The reimbursement applies only to the reliability component and is capped at \$105 nominal dollars per kW in 2026 (California Independent System Operator, 2025); delivery network upgrades are reimbursed only to the extent that the generator is awarded transmission plan deliverability (California Independent System Operator, 2024).

⁶Some socialized-cost systems include ongoing transmission charges that vary by location, aiming to recover network costs from generators in proportion to their use. These charges often have high year-to-year volatility and are subject to regulatory caps that limit how strong the locational signal can be, features that likely undermine their value as long-run investment signals (Office of Gas and Electricity Markets, 2023).

ratepayers bear the costs. In practice, network expansion in socialized-cost models, whether planned proactively or triggered by connection requests, requires regulatory approval that weighs its cost to ratepayers. When connection requests outpace this buildout, scarce network capacity is rationed primarily by waiting rather than by price (European Commission, 2025).

8 Conclusion

We use data from PJM to study how grid connection costs affect investment in generation capacity. We find that network upgrade costs cause planned generators to withdraw from the queue. We also find that generators are less likely to face high costs in areas with recent transmission investment, even though such investment rarely targets new generation. This finding suggests that the rise in network upgrade costs over the last decade reflects a mismatch between the pace of transmission investment and the growing demand for transmission capacity.

A key outstanding question is how much reducing network upgrade costs would increase generator completion. One extreme is that the conditional probability of advancing in the queue for projects with network upgrade costs would increase to that of those without these costs, resulting in a large increase in completion. The other extreme is that developers would plan fewer projects, such that the same amount of capacity would be completed. There would still be significant cost savings even in the second case: developers would no longer pay to develop projects that are later canceled due to high grid connection costs. Lower costs would also reduce demand for the necessary studies and hence delays; survey and empirical evidence show that delays in these studies lead to project cancellation (Silverman et al., 2024; Johnston, Liu and Yang, 2026).

Our finding that network costs are lower after transmission spending has implications for the current debate over transmission cost allocation. It suggests that regional transmission spending paid for by load can reduce interconnection costs, i.e., this investment provides value to connecting generators, who should therefore bear some of its cost. Economic theory suggests that, if those paying for the transmission infrastructure do not capture all of its benefits, transmission investment will be insufficient (Davis, Hausman and Rose, 2023).

These findings also have implications for transmission planning. Many generators with high network upgrade costs leave the queue after the second study; historically, such generators never entered the planning process. FERC’s 2024 transmission planning reform (Order 1920) addresses this gap by requiring grid operators to consider network upgrades that have been repeatedly identified in the interconnection process (Federal Energy Regulatory Com-

mission, 2024); it also permits, but does not require, grid operators to incorporate zones suitable for large amounts of new generation into long-term planning. Some grid operators are going further: FERC recently approved SPP’s consolidated planning process, in which SPP proactively plans transmission for connecting generators and charges them a fixed, up-front per-MW fee for connection (Southwest Power Pool, 2025; Federal Energy Regulatory Commission, 2026*b*). Our findings suggest that reforms like SPP’s, which go beyond the minimum required by Order 1920, may be essential for achieving reliability and climate goals.

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During the preparation of this work the author(s) used Chat GPT and Claude in order to assist with extracting substation information from transmission investment descriptions, editing code, and improving the clarity and readability of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Appendix

A Additional Data and Analysis Details

A.1 Interconnection Cost Variables

We collect data from each of the three engineering studies introduced in Section 2.1.1: the feasibility study (first study), the system impact study (second study), and the facilities study (third study). Studies are sometimes revised, and we record data from the final version of each study (as of August 2024). For projects on PJM’s expedited path, a single combined feasibility-and-impact study substitutes for the first two studies. We classify it as the first study, and these projects may or may not subsequently receive a third study.

When collecting interconnection cost data, we follow PJM’s categorization and differentiate between two types of costs: POI costs and network upgrade costs. The POI cost, also known as the physical connection cost, is the estimated cost to connect a generator to the electric grid. In the engineering studies, these expenses are classified as the direct connection cost, non-direct connection cost, and attachment cost. We calculate the POI cost as the sum of these three components. We record the POI cost as zero if a study explicitly provides a zero-cost estimate or states that no such costs were identified. After applying this rule, 28, 17, and 10 percent of observations are missing POI costs for the first, second, and third studies. For our analysis, we replace these missing values with zero, so that every observation in our sample has a non-missing POI cost.

The network upgrade cost is the estimated cost for transmission system enhancements that the generator is responsible for paying. We extract these costs from sections of the engineering studies that are labeled as referring to network upgrade costs. Network upgrade cost estimates are rarely provided in the first study: only 4 percent of observations have a non-missing network upgrade cost for the first study. Network upgrade costs may also be missing in the second and third studies. If a study explicitly reports zero costs or states that no such costs were identified, we record the network upgrade cost as zero. After applying this rule, 42 percent of second-study observations and 39 percent of third-study observations are missing network upgrade costs. These observations do not mention network upgrade costs anywhere in their engineering studies, nor do they include a section relevant to these costs. We assume this lack of information means the project was not responsible for any contribution to network upgrade costs and replace these missing values with zero.

When multiple projects contribute to a grid overload, PJM allocates upgrade costs in proportion to each project’s contribution. The combined cost is the total cost of upgrades

to which a project may contribute and therefore provides an early indication of its network upgrade cost. Because this estimate is often reported in the first study before cost shares are resolved, we use it as our first-study measure of network upgrade cost. Following the same approach as with the network upgrade cost, we record the combined cost as zero if a study explicitly reports a zero combined cost or states that no such costs were identified. After applying this rule, 43 percent of first-study observations have a missing combined cost. These observations have no mention of a combined cost in their studies, so we assume their combined cost is zero.

To illustrate these cost categories, Figure A1 shows an example of how costs appear in the first study. The first table lists the costs AD1-151 is solely responsible for: its attachment facilities and the direct and non-direct connection network upgrades dedicated to its own interconnection. These are the three components of the POI cost defined above and total \$9.1 million. The second table lists the network upgrades AD1-151 may instead share with other projects; their \$281.07 million total is the project's combined cost. Because these upgrades are shared, AD1-151 is responsible for only a portion of the \$281.07 million, and later studies report a network upgrade cost that reflects its allocated share of the combined cost. The combined cost therefore tends to be much larger than the network upgrade costs reported in the second and third studies.

Figure A1: Example cost breakdown for the first study

The AD1-151 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 1,800,000
Direct Connection Network Upgrades	\$ 6,300,000
Non Direct Connection Network Upgrades	\$ 1,000,000
Total Costs	\$ 9,100,000

In addition, the AD1-151 project may be responsible for a contribution to the following costs:

Description	Total Cost
New System Upgrades	\$ 0
Previously Identified Upgrades	\$ 281,065,000
Total Costs	\$ 281,065,000

Notes: The first block reports the costs charged to AD1-151 alone; the second block reports the shared upgrades it may help fund, which we record as its combined cost.

We construct a measure of project capacity that we use to transform these costs into the cost per kW measures used in our analysis. Each request for interconnection specifies the requested kW of both energy resource and capacity resource. Energy resource is the kW that

can participate in the energy market; capacity resource is the kW that can participate in the capacity market. These two values are highly correlated, though the requested capacity kW tends to be lower. As our measure of capacity, we use the maximum of these two values.

A.2 Analysis with Transmission Owner (TO)–Owned Generation Capacity

PJM does the interconnection studies in conjunction with the local TO. Some TOs own generation assets, which could create a conflict of interest. New generators typically have lower marginal costs of production, which may suppress electricity prices and reduce revenues for incumbents, including those owned by TOs.

To test whether network upgrade costs are higher when the TO owns incumbent generation, we use publicly available EIA Form 860 data to construct a measure of the TO-share of generation. Form 860 reports each generator’s nameplate capacity, owner, and the owner of the transmission system to which its plant is interconnected. We manually match transmission system owners and generator owners to PJM TOs. For each TO and year, total interconnected capacity is the sum of nameplate capacity across operable generators, and TO-owned capacity is the portion owned by that TO. The TO-share of generation is the ratio of TO-owned capacity to total interconnected capacity.

Twelve of the twenty-one transmission owners in our sample own no generation in their service territory; among the nine that do, the mean share of in-territory generating capacity they own is 13 percent, with AEP having the highest share at 36 percent.

A.3 Transmission Spending Analysis Variables

We use data from PJM’s Regional Transmission Expansion Plan (RTEP) to construct a set of transmission spending variables. These data are publicly available from PJM and include all transmission projects that were identified by or proposed to PJM from 2003 to 2024.

These spending measures capture transmission spending that is near the generator receiving the second study. The RTEP project descriptions list the affected substations, and these are the locations to which we assign transmission spending. For each generator, we create four sets of transmission spending variables by aggregating projects in geographic proximity to the generator’s POI substation. The first three sets of measures aggregate total investment at substations located within 10km, 20km, and 50km of the POI. PJM reports a single cost for each RTEP project rather than the costs attributable to individual substations. We therefore include the full cost of an RTEP project in our spending measure if any of its substations falls within the relevant radius of the generator’s POI.

For the fourth measure, we define regions by clustering substations based on the similarity of their local wholesale electricity prices and location. Each generator is then assigned the transmission investment associated with its corresponding region. Wholesale prices are measured as the average of hourly real-time locational marginal prices (LMPs) at each substation, using data from the 5th, 15th, and 25th of each month over the study period. We apply a hierarchical clustering algorithm that groups substations into 50 regions based on two dissimilarity measures: geographic proximity and wholesale market prices. Both measures are normalized by their maximum values to ensure comparability. We assign equal weights to LMP similarity and spatial distance. Our results are robust to alternative specifications, including increasing the LMP similarity weight to 60 percent and adjusting the number of regions to 40 or 60.

More precisely, let p_i denote substation i 's average real-time LMP and (x_i, y_i) its coordinates. We form two dissimilarity matrices, $D_{ij}^P = |p_i - p_j|$ in price space and $D_{ij}^G = \|(x_i, y_i) - (x_j, y_j)\|$ in geographic space, and normalize each by its maximum entry, $\tilde{D}^P = D^P / \max D^P$ and $\tilde{D}^G = D^G / \max D^G$. For a partition of the substations into K regions $\mathcal{P}_K = \{C_1, \dots, C_K\}$, let

$$W(\tilde{D}, \mathcal{P}_K) = \sum_{k=1}^K \frac{1}{2|C_k|} \sum_{i,j \in C_k} \tilde{D}_{ij}^2$$

denote the pseudo within-cluster inertia under a dissimilarity matrix $\tilde{D}$. Define the weighted homogeneity criterion

$$Q_\alpha(\mathcal{P}_K) = (1 - \alpha) W(\tilde{D}^P, \mathcal{P}_K) + \alpha W(\tilde{D}^G, \mathcal{P}_K), \quad \alpha = 0.5.$$

We form regions using a Ward-type hierarchical algorithm with a spatial constraint (Chavent et al., 2018), which merges clusters to minimize the increase in Q_α at each step. We cut the resulting dendrogram into $K = 50$ regions. Setting $\alpha = 0.5$ weights price similarity and geographic proximity equally.

When constructing these measures, we include only RTEP investments with operation dates between three years before and one year after the study issue date. We include projects that are within a year of their operation date because these are often accounted for in the estimated network upgrade costs. We do not include projects that began operation more than three years before because we expect other generators to have entered in the interim and eroded the benefits from these older investments. For the same reason, we include total entry in the geographic area corresponding to the spending variables as a control variable in these regressions. Our measure of entry is the log of the MW that entered the PJM queue in that geography in the three years before the study issue date.

Finally, from each transmission spending measure, we create four indicator variables that capture different levels of spending. These bins are below the 50th percentile, the 50th–75th percentile, the 75th–90th percentile, and above the 90th percentile. In the analysis, we include the top three bins, so coefficients should be interpreted relative to spending levels below the median.

B Additional Figures and Tables

Figure B6 shows that the probability of receiving a positive network upgrade cost differs significantly across TOs but shows no clear relationship with the share of TO-owned generation capacity. Some TOs with no TO-owned generation have relatively high probabilities of a positive network upgrade cost.

In Figure B10, we plot PJM’s transmission investment (RTEP) across years. We also separate two types of RTEP. The red represents baseline investment, which is necessary to maintain the regional grid reliability. The green represents the supplemental investment, which addresses local operational and economic needs. The overall level of investment has been stable over time, but the share of baseline investment is lower in the second half of the sample.

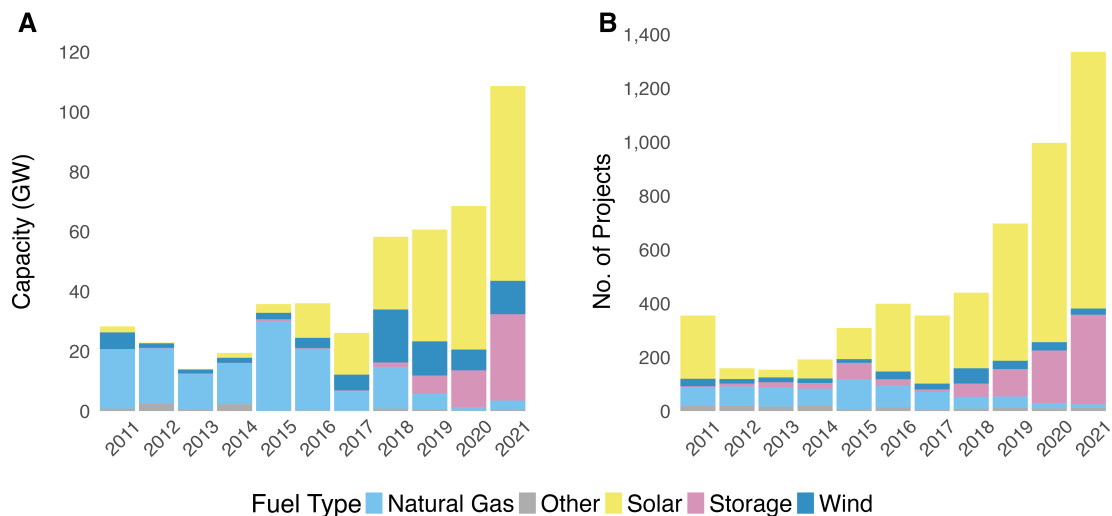


Figure B1: **Entrants to the PJM Queue by Year and Fuel Type.** New entrants to the PJM queue 2011–2021. (A) Total Capacity in GW. (B) Number of Entrants. Storage is standalone storage; hybrid storage projects are counted as wind or solar.

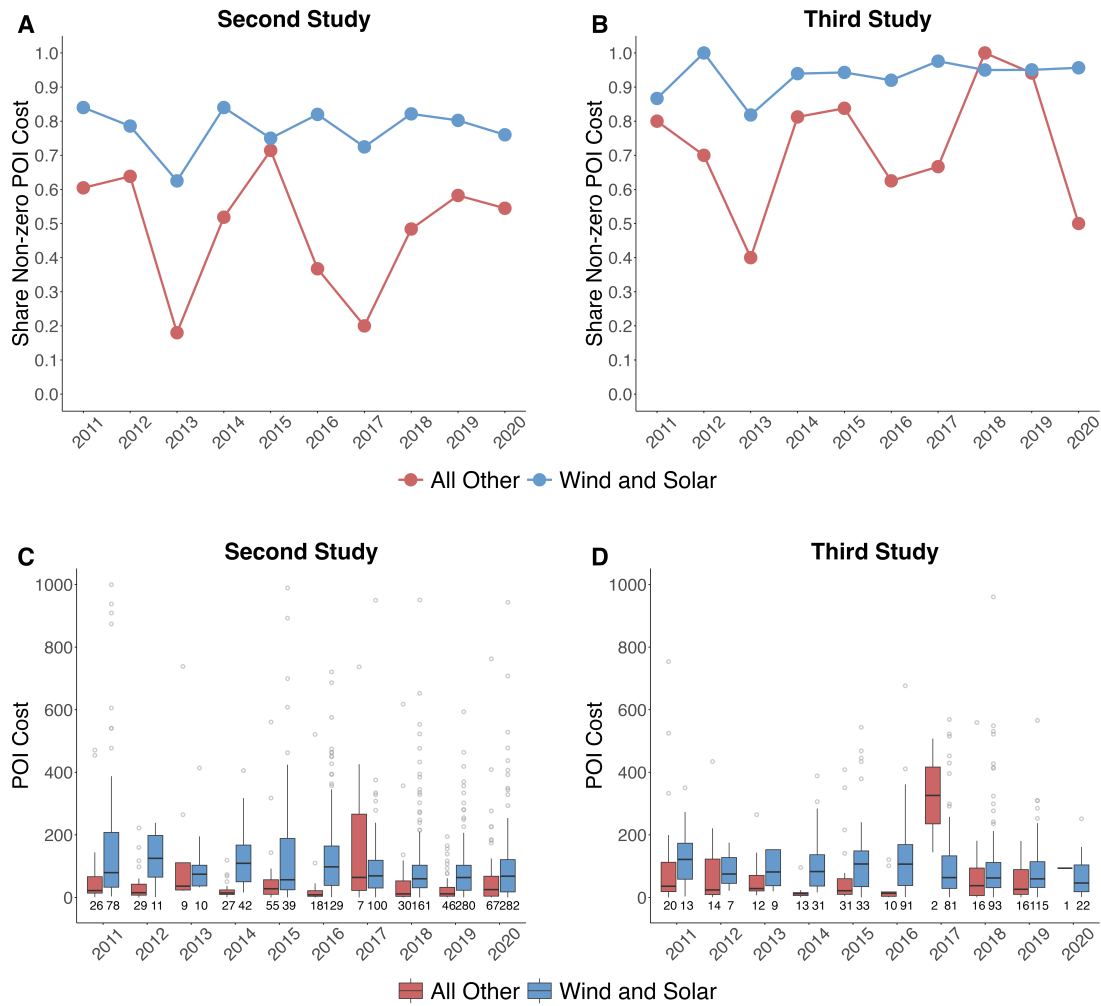


Figure B2: **Point of Interconnection (POI) Costs by Year of Queue Entry and Fuel** (A) Share of projects with nonzero POI costs in the second study. (B) Share of projects with nonzero POI costs in the third study. (C) Box plot of second study POI costs in \$/kW for projects with nonzero costs. (D) Box plot of third study POI costs in \$/kW for projects with nonzero costs.

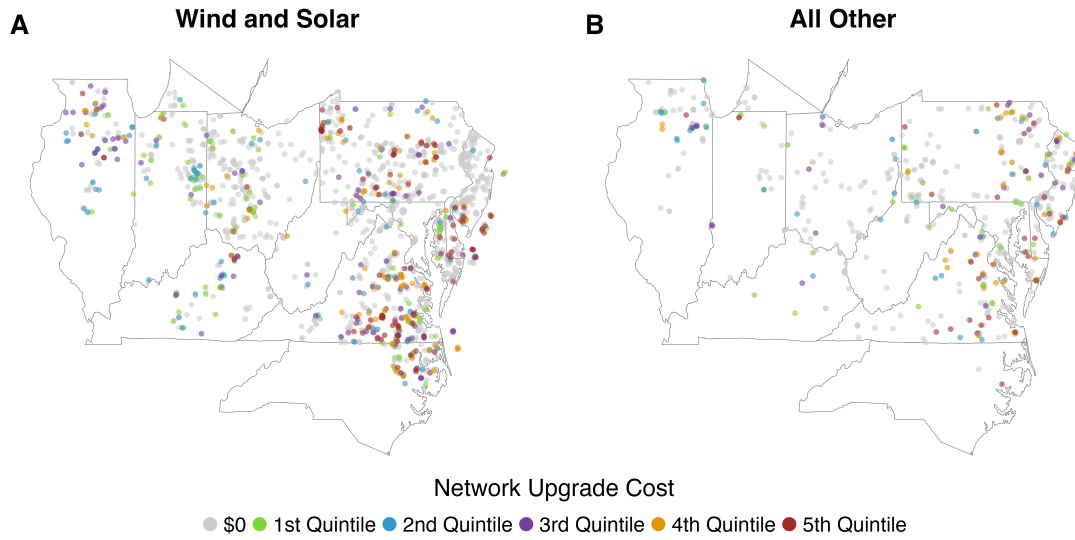


Figure B3: **Network Upgrade Costs by Generator Fuel Type** Estimated network upgrade costs from the second study for 1,447 wind and solar projects (**A**) and 619 projects of all other fuel types (**B**).

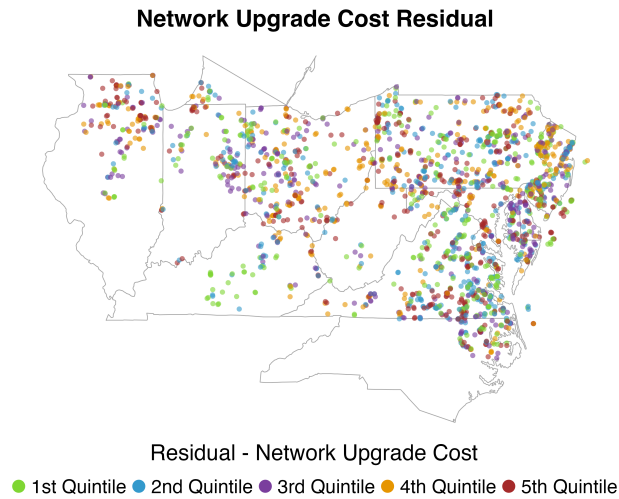


Figure B4: **Network Upgrade Cost Residual** Residual variation in network upgrade costs from the second study for 2,066 projects that entered the queue in 2011–2020. We regress network upgrade costs on project size (4 bins), fuel types (solar, wind, storage, other), the interactions of size bins and fuel types, transmission zone, high-voltage, whether the project is a capacity increase at an existing plant, and queue year. We then plot the residuals.

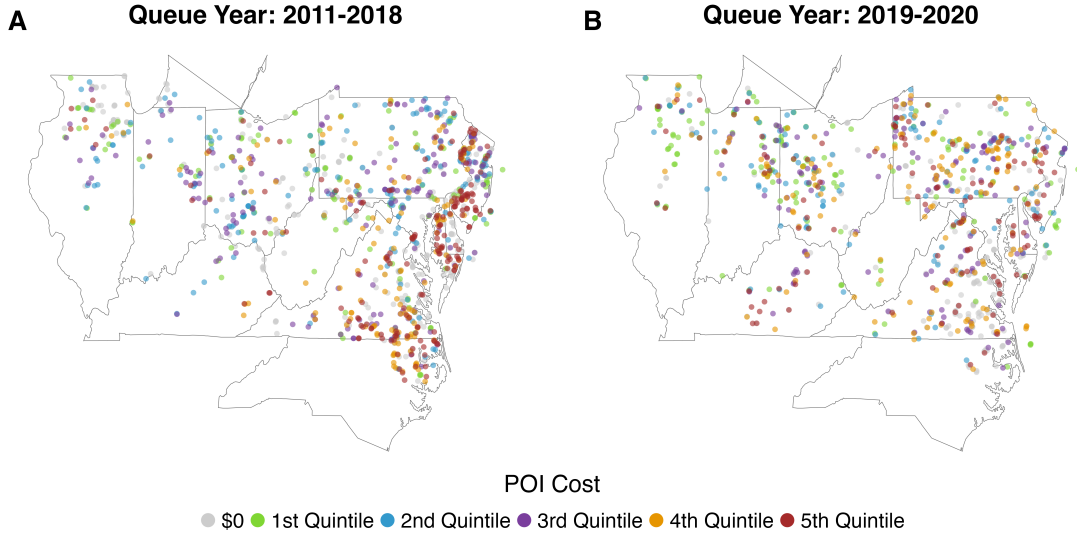


Figure B5: **Point of Interconnection (POI) Costs by Year of Queue Entry and Location** Estimated POI costs from the second study for the 1,144 projects that entered the queue in 2011–2018 (A) and the 922 projects that entered the queue in 2019–2020 (B). POI costs are grouped into five bins defined by the 20th, 40th, 60th, 80th, and 100th percentiles, in addition to a zero-cost group. The corresponding cutoffs are 14, 38, 77, and 136 \$/kW.

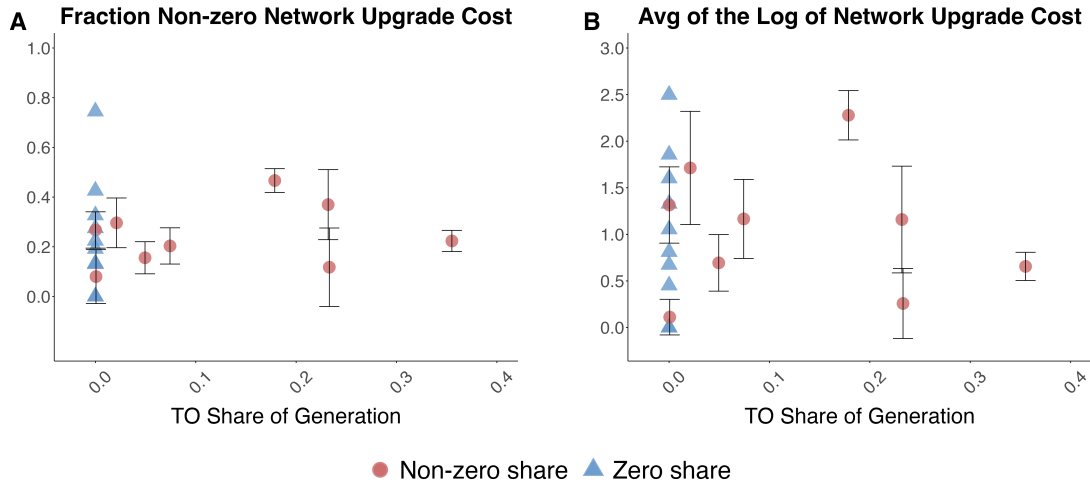


Figure B6: **Network Upgrade Costs and Share of Transmission-Owner (TO) Generation Capacity** A red dot is a TO with positive share of TO-owned generation capacity. A blue triangle is a TO with no TO-owned generation capacity. In (A), the y axis plots, by TO, the mean of an indicator for nonzero network upgrade cost. In (B), the y axis plots, by TO, the average of the log of network upgrade cost. For TOs with a positive share of TO-owned capacity, we also plot the corresponding 95 percent confidence intervals. The sample includes projects queued 2011–2020 ($N = 2,066$).

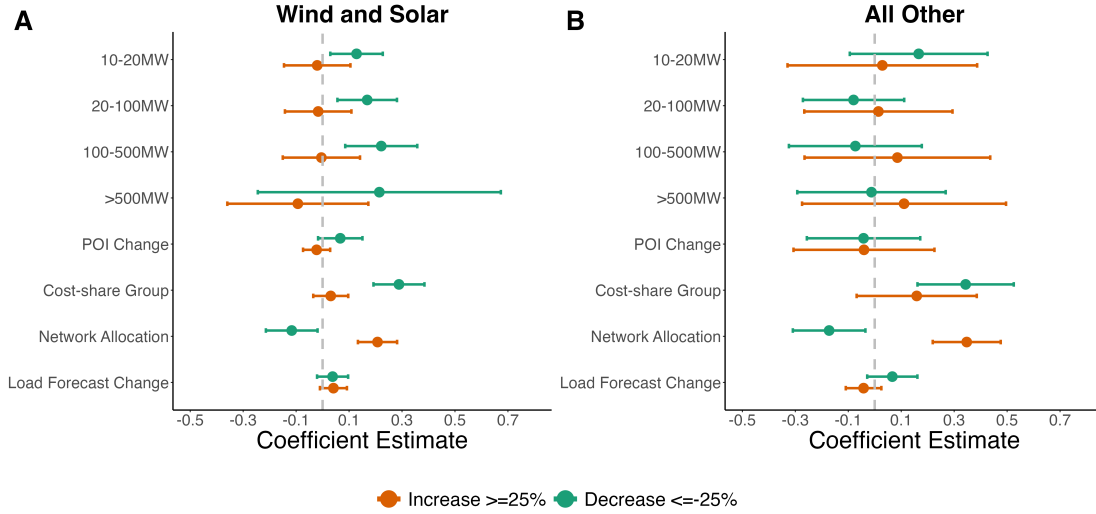


Figure B7: Predictors of Large Increases versus Decreases in Network Upgrade Costs Between the Second and Third Study. The figure reports estimated coefficients from two linear probability models, each estimated on projects with a third study and queue year 2011 or later. Orange circles report coefficients from a model that predicts the probability of a network upgrade cost increase of 25 percent or more between the second and third study (sample mean is 0.08 for Wind and Solar and 0.2 for All Other). Green circles report coefficients from a model that predicts the probability of a network upgrade cost decrease of 25 percent or more between the second and third study (sample mean is 0.14 for Wind and Solar and 0.16 for All Other). Size coefficients are relative to an omitted bin for 0–10 MW. POI Change indicates the POI changed between the second and third studies. Cost-share group is an indicator for a second study that lists other generators that a project shares network upgrade costs with. Network allocation is an indicator for an explicit network upgrade cost in the second study; this cost may be \$0. Load Forecast Change is the signed change, measured in GW, in the five-year-ahead PJM peak-load forecast for the project’s transmission owner between the second and third studies. The model controls for fuel type, whether the project is a capacity increase at an existing plant, year of entry into the queue, state, transmission owner, and voltage.

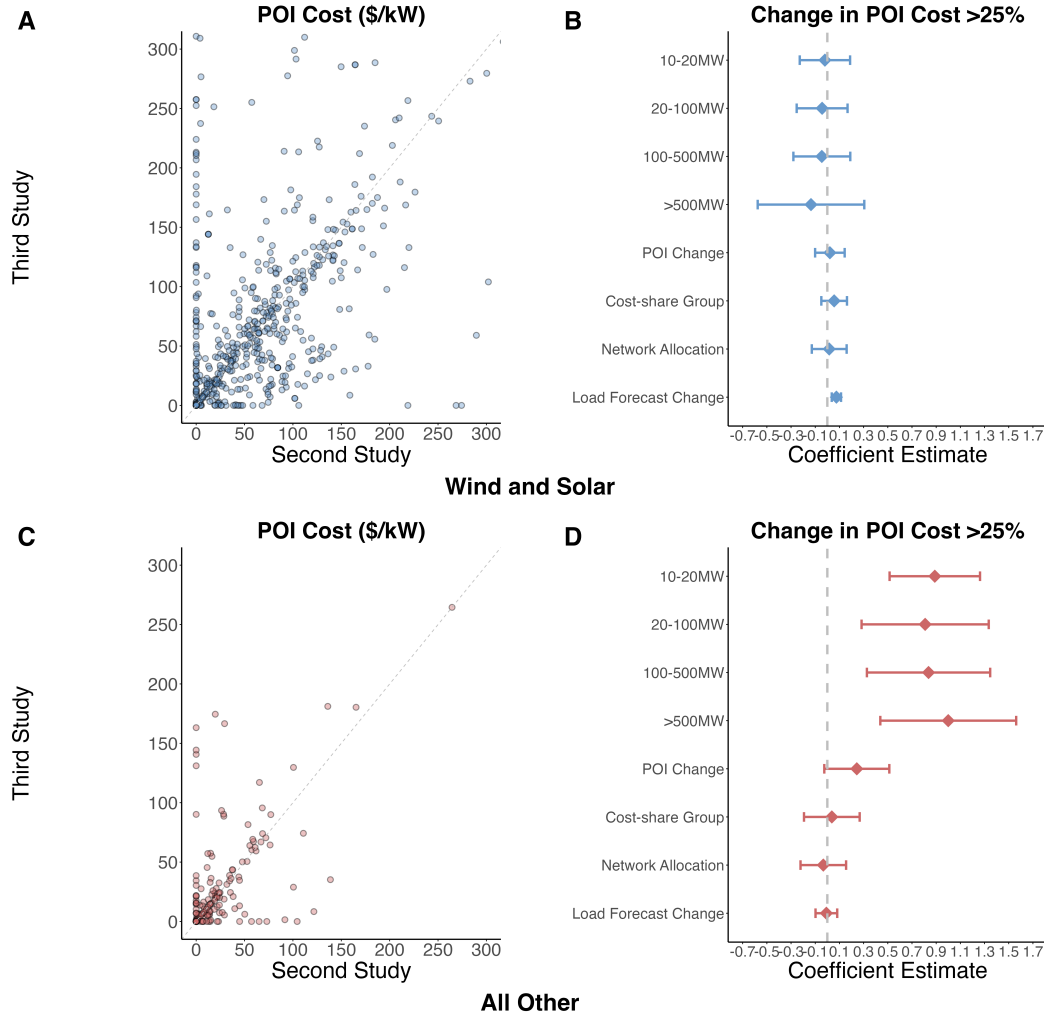


Figure B8: **Comparison of Second- and Third-Study Point-of-Interconnection (POI) Costs for the Same Project and Predictors of Large Changes, by Fuel Type** Panels (A) and (C) plot third-study versus second-study POI costs for projects observed in both studies, separately for wind and solar (Panel A: 573 projects; 62 projects with at least one cost greater than \$300/kW are excluded from the plot) and all other fuel types (Panel C: 199 projects; 9 projects with at least one cost greater than \$300/kW are excluded). Panels (B) and (D) report estimated coefficients from a linear probability model predicting the probability of a POI cost change greater than 25 percent between the second and third study, estimated separately for wind and solar (Panel B, sample mean = 0.62) and all other fuel types (Panel D, sample mean = 0.70). Size coefficients are relative to an omitted bin for 0–10 MW. POI Change indicates the POI changed between the second and third studies. Cost-share group is an indicator for a second study that lists other generators that a project shares network upgrade costs with. Network allocation is an indicator for an explicit network upgrade cost in the second study; this cost may be \$0. Load Forecast Change is the absolute change, measured in GW, in the five-year-ahead PJM peak-load forecast for the project’s transmission owner between the second and third studies. The model controls for fuel type, whether the project is a capacity increase at an existing plant, year of entry into the queue, state, transmission owner, and voltage.

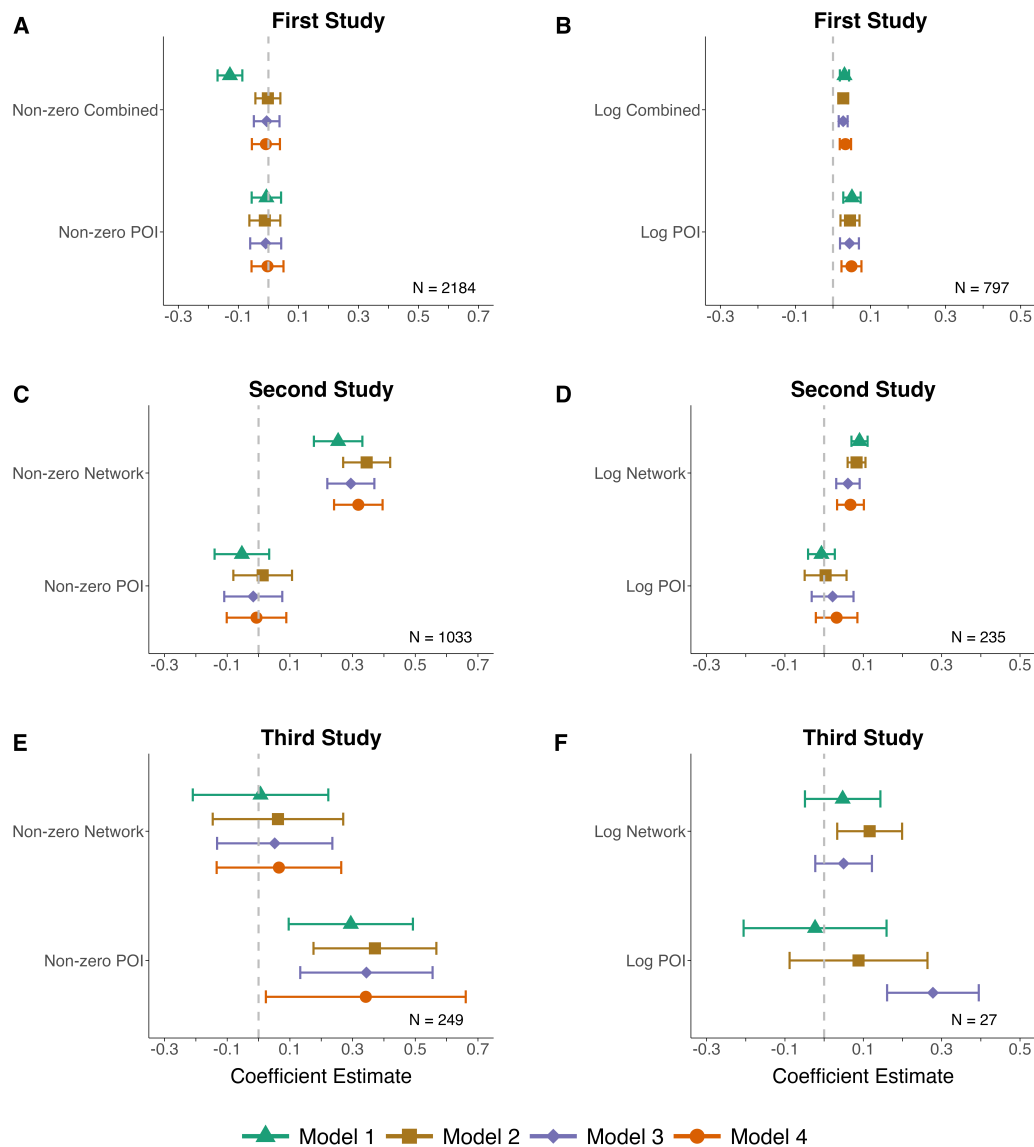


Figure B9: **Effect of Interconnection Costs on Probability of Withdrawal: Wind and Solar Projects.** Coefficient point estimates (triangles, squares, diamonds, and circles) and 95 percent confidence intervals from a linear probability model estimated on wind and solar projects only. Projects that entered the queue 2011–2020, active projects excluded. Dependent variable is an indicator for withdrawing from the queue before the next study (first and second study) or beginning operation (third study). The means of the dependent variables are 0.288 (A), 0.231 (B), 0.448 (C), 0.617 (D), 0.470 (E), and 0.481 (F). Model 1 (green triangles) has no controls, Model 2 (brown squares) controls for project size and fuel type, Model 3 (purple diamonds) also controls for year of entry into the queue, and Model 4 (orange circles) also controls for state, transmission owner, and voltage. Panels A, C, and E plot coefficients on indicators for a nonzero combined or network upgrade cost (the extensive margin), estimated on all wind and solar projects that receive the first, second, and third study, respectively. Panels B, D, and F plot coefficients on the log of the cost (the intensive margin) among wind and solar projects with positive costs in that study. The samples are accordingly: all wind and solar projects with a first study (A), those with positive first-study combined and POI costs (B), all with a second study (C), those with positive second-study network and POI costs (D), all with a third study (E), and those with positive third-study network and POI costs (F).

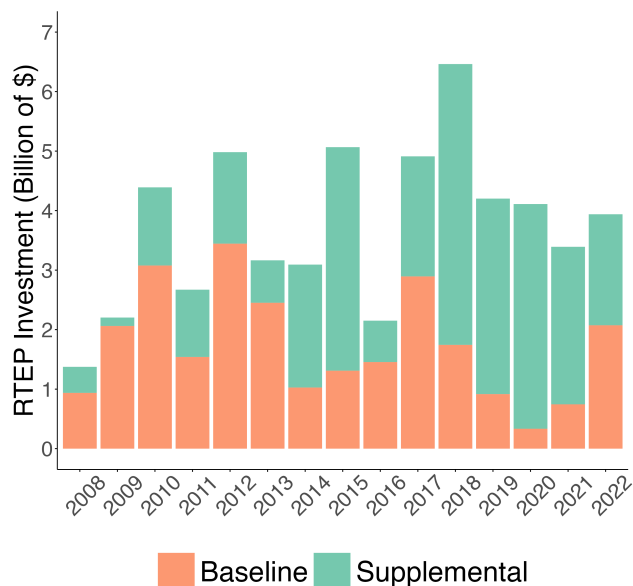


Figure B10: **Transmission Investment in PJM by Regional Transmission Expansion Plan (RTEP) Year.** Total spending on transmission investments in PJM’s RTEP. Transmission projects are assigned to the year they entered the process—either the date the project was first reviewed by the Transmission Expansion Advisory Committee or its in-service date, whichever is earlier. Baseline projects are identified by PJM to address regional reliability or operational needs, and supplemental projects are proposed by individual transmission owners to address local requirements.

Table B1: Predictors of Network Upgrade Costs and POI Costs

Dependent variable	Network Upgrade			POI		
	Nonzero	Cost in \$/kW	Log of Cost	Nonzero	Cost in \$/kW	Log of Cost
Project Size						
... (10MW, 20MW]	0.10*** (0.03)	43.43* (25.04)	-0.57* (0.34)	0.04 (0.04)	-85.09*** (19.75)	-0.91*** (0.16)
... (20MW, 100MW]	0.25*** (0.03)	85.55*** (24.90)	-0.72** (0.32)	0.24*** (0.03)	-66.79*** (19.20)	-0.94*** (0.16)
... (100MW, 500MW]	0.42*** (0.04)	125.47*** (28.30)	-0.71** (0.34)	0.28*** (0.04)	-98.75*** (20.10)	-1.45*** (0.18)
... >500MW	0.60*** (0.06)	134.48*** (31.01)	-0.75 (0.51)	0.43*** (0.05)	-125.93*** (23.66)	-2.03*** (0.25)
Fuel Type						
... Solar	0.03 (0.04)	-9.74 (25.05)	-0.22 (0.44)	0.27*** (0.04)	22.59* (12.52)	0.51** (0.21)
... Wind	0.11** (0.05)	45.41* (27.00)	-0.17 (0.40)	0.21*** (0.04)	9.01 (11.38)	0.32 (0.23)
... Storage	0.10** (0.05)	18.82 (30.23)	-0.54 (0.47)	0.29*** (0.05)	-14.87 (14.15)	-0.14 (0.25)
... Other	0.02 (0.05)	12.84 (18.97)	0.19 (0.73)	-0.00 (0.06)	17.57 (17.88)	1.23** (0.53)
Transmission Zone						
... MidAtlantic	-0.17** (0.07)	-134.71** (59.49)	-0.57 (0.69)	0.19*** (0.07)	-27.42 (26.02)	-0.31 (0.39)
... SouthWestAtlantic	-0.19*** (0.06)	-137.37*** (45.13)	-2.52*** (0.81)	0.13 (0.09)	-16.79 (31.39)	-0.03 (0.38)
... Southern	0.19** (0.09)	13.34 (63.15)	0.71 (0.56)	0.07 (0.07)	-24.56 (28.85)	0.02 (0.42)
... Western	-0.10 (0.06)	-76.24 (48.50)	-0.90 (0.59)	0.24*** (0.06)	-19.67 (27.27)	-0.46 (0.36)
High Voltage	-0.06** (0.03)	-46.76** (18.56)	-0.14 (0.19)	-0.02 (0.03)	-11.21 (7.24)	-0.13 (0.09)
Capacity increase at existing plant	-0.03 (0.03)	32.38 (22.59)	0.45** (0.21)	-0.42*** (0.03)	-71.46*** (7.11)	-2.39*** (0.19)
Mean dependent var.	0.30	90.00	4.32	0.71	71.68	3.69
R ²	0.24	0.15	0.41	0.40	0.19	0.44
N	2,066	2,066	628	2,066	2,066	1,457

Generators that entered the queue 2011–2020 and received a second study. SEs in parentheses; clustered by substation. Omitted fuel is natural gas, omitted zone is East Mid-Atlantic. High Voltage is an indicator for voltage >115kV. All specifications control for state and the year of entry to the queue. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B2: Summary Statistics for PJM Transmission Planning Data

	Baseline RTEP		Supplemental RTEP	
	Mean	Std. Dev.	Mean	Std. Dev.
Investment (in Million \$)	11.09	39.69	10.79	24.38
No. of Substations	1.38	0.67	1.48	0.78
Voltage (kV)	199.82	123.29	148.37	105.53
Time to In-Service (Months)	39.13	20.60	36.25	26.58
N	2,627		2,500	

The sample includes all PJM Regional Transmission Expansion Plan projects that went into service or are proposed to be in service 2008–2026. Projects are classified by PJM as either baseline or supplemental. Investment refers to the estimated cost of the project in million dollars. The number of substations represents the total substations associated with the project’s construction. Voltage indicates the operating voltage of the transmission asset. Time to In-Service measures the number of months from the initial planning date to the in-service date.